

BMS Integration Specification Document

Smart Building Platform – Revision 2.1, March 15, 2026

1. Scope of Application

This document specifies the integration between a Building Management System (BMS) and a cloud platform. The BMS is responsible for monitoring and controlling the building's HVAC systems, lighting, and power meters.

2. System Architecture

- Air Handling Unit (AHU) Controller, Model FX-PCG2611-0
- Variable Air Volume (VAV) Terminal Controller
- Occupancy sensor and CO2 sensor (0~2,000 ppm)
- Edge device and gateways (BACnet/IP, Modbus TCP, MQTT)

3. Network Requirements

Item	Values
BMS server	192.168.1.100
BACnet/IP port	47808
MQTT broker	mqtt.example.com:8883
Polling interval	15 minutes

4. Alarm handling

When an alarm is triggered for a chiller, the system notifies the maintenance team via email (support@example.com) and creates a work order. If the alarm is not confirmed within 30 minutes, it is escalated to the facility manager.

Note: Any change to the setpoint requires approval from the operator.

5. Energy management

The Energy Management System (EMS) records energy consumption every 15 minutes. By implementing demand response and adjusting the air conditioning setpoint from 24°C to 26°C, peak demand was reduced by 18% by the summer of 2025.

6. Data retention

The data from trend logs is stored on the cloud platform (<https://bms.example.com/trends>) for 13 months.

